

2(a)	<i>any four from:</i> 1 population / a species, show(s) genetic, variation / diversity ; 2 <i>ref. to</i> chance / random / spontaneous, mutation ; 3 <i>ref. to</i> selection pressure / competition / predation / disease ; 4 <u>fitter</u> / <u>fittest</u> OR better <u>adapted</u> / best <u>adapted</u>, (individuals) survive ; ora 5 reproduce / breed, <u>more</u> OR have <u>more</u> offspring ; ora 6 pass on, beneficial / advantageous / favourable / desirable, <u>allele(s)</u> ; ora 7 change / increase, in <u>allele</u> frequency ;	4
2(b)(i)	<i>any two from:</i> <u>normal distribution</u> ; <u>5</u> is, usual / commonest / most frequent / peak, <u>clutch size</u> ; few(er) clutches, are very large and very small / are of size 2 and 8 / lie at (both) extremes ;	2
2(b)(ii)	<i>any four from:</i> 1 <u>stabilising selection</u> ; 2 4 / 5 (eggs) or mean / average (clutch size), is optimum / better / best / selected for / has selective advantage / gives most (surviving) offspring ; 3 low clutch size / 2 / 3 , gives few(er) (surviving) offspring ; 4 high clutch size / 6 / 7 / 8 , not all offspring survive ; 5 high clutch size <i>linked to</i> more competition (between chicks) for, food / parental care ; 6 high clutch size <i>linked to</i> (more) predation (of eggs / chicks) ; 7 high clutch size <i>linked to</i> (more) disease (of eggs / chicks) ;	4

Question	Answer	Marks
2(c)(i)	<i>step 1:</i> $17 / 86$ or $\underline{0.198} / 0.20 (q^2) ;$ <i>step 2:</i> $\sqrt{0.198}$ or $\underline{0.445} / 0.45 / 0.44 (q)$ AND $1 - 0.445$ or $\underline{0.555} / 0.55 / 0.56 (p) ;$ <i>step 3:</i> $2 \times 0.555 \times 0.445$ or $0.49(395) / 0.495 (2pq)$ AND $0.49(395) \times 86$ or $42 / 42.479 / 42.48 / 42.5 / 42.57 ;$	3
2(c)(ii)	<i>any two from:</i> 1 small population ; 2 (natural / artificial / sexual) selection ; 3 non-random mating / inbreeding ; 4 migration (of individuals) ; 5 (new) mutation (is occurring); 6 non-diploid organisms ; 7 overlapping generations ; 8 asexual reproduction ; 9 allele frequencies not equal in both sexes ;	2